Decoupled ignition and propagation benchmarking for operational wildfire nowcasting

A reproducible computational framework — not a new boosting algorithm

1 | SEPARATE THE TARGETS

Primary ignition

fire at (i,t) without t-1
fire within 25 km

Tactical propagation

fire at (i,t) within
25 km of a t-1 fire

No persistence leakage

Same fire, two tasks.
D2 keeps the labels disjoint.

2 | GATE AND DECOUPLE

Inputs

VIIRS · ERA5-Land · MapBiomas · history

Biophysical gate

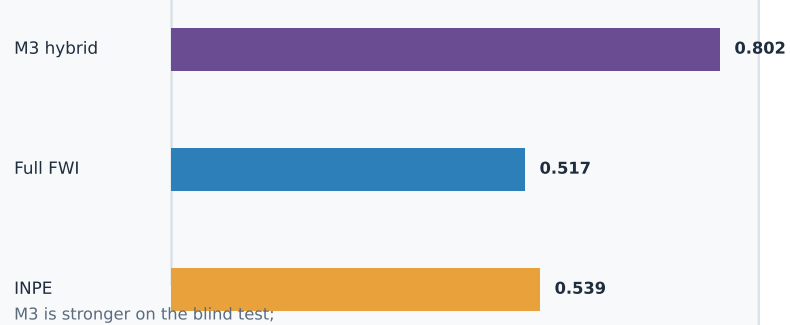
M3 hybrid
(no t-1 fire)

M1 fallback
(no history)

Causal t-1 propagation
+ frozen evaluation harness

3 | BLIND BENCHMARK

ROC-AUC | Chaco | retrospective 2024–2025



M3 is stronger on the blind test;
M1 is the fallback for unseen cells.

D2 targets · gated M3/M1 · causal propagation · open benchmark